

MindMap: An EEG and AI-Driven Study for Mental Health Assessment and Categorization

1st Md. Takoat Hossain

Electrical And Electronic Engineering
Shahjalal University of Science and Technology
Sylhet, Bangladesh
takoathossain.com

3rd Dr.Md. Rasedujjaman

Electrical And Electronic Engineering
Shahjalal University of Science and Technology
Sylhet, Bangladesh
mrased-eee@sust.edu

2nd Shirajul Abedin Shimul

Electrical And Electronic Engineering
Shahjalal University of Science and Technology
Sylhet, Bangladesh
shimul122000@gmail.com

4th Md. Shariful Islam

Electrical And Electronic Engineering
Shahjalal University of Science and Technology
Sylhet, Bangladesh
shariful-eee@sust.edu

Abstract—Major Depressive Disorder (MDD) is a major health challenge worldwide, but clinical diagnosis still relies on subjective assessment tools that lack a physiological basis. In this paper, we propose MindMap, a diagnostic framework based on Spatial-Temporal Graph Attention Networks (ST-GAT) in electroencephalogram (EEG) signals for automatic MDD categorization. The framework is initialized by transfer learning on the SEED-IV emotion recognition dataset achieving 46.30% on a four-class task. Clinical fine-tuning on the resting-state MODMA dataset achieves 68.47% binary diagnostic accuracy (MDD versus healthy control) with AUC of 0.73. The independent population level validation on 294 subjects resulted in an accuracy of 71.79% with a standard deviation of 7.66%. The statistical analysis of the graph attention weights revealed the most significant ($p < 0.001$) Frontal-Midline connectivity between FP1 and Fz electrodes in the Delta and Theta bands, while individual channel power at the same site is non-significant ($p = 0.1065$), validating the graph-based formulation over channel-wise alternatives.

Index Terms—Electroencephalography, Major Depressive Disorder, Graph Attention Networks, Transfer Learning, Biomarker Discovery, Mental Health Assessment, Cross-Device Validation.

I. INTRODUCTION

According to the World Health Organization [1], Major Depressive Disorder is the leading cause of years lived with disability and an estimated 280 million people worldwide suffer from it [1]. However, even this is dependent on structured psychiatric interviews and self-report instruments like the Hamilton Rating Scale and the Beck Depression Inventory for clinical diagnosis. These tools are known to be sensitive to clinician training, patient willingness to disclose, and the cultural framing of emotional distress. Currently, no physiological measurement is included in standard diagnostic criteria.

EEG provides a non-invasive, high temporal resolution analysis of neural activity and has been explored as a potential depression biomarker for many decades [2]. But traditional analysis methods of EEG data treat each electrode independently and do not consider the relational structure of brain activity. Neuro scientific findings confirm MDD is a disorder

of neural connectivity, not a localized dysfunction [3]. We have seen disrupted prefrontal-limbic communication, decreased frontal-midline coherence and atypical posterior alpha activity in populations with MDD, but standard spectral and machine learning pipelines do not allow for direct modeling of these inter-electrode dependencies.

Graph neural networks naturally provide a solution by considering the relations of the electrodes as edges in a learnable graph. We propose a framework called MindMap based on Spatial-Temporal Graph Attention Networks (ST-GAT) which considers 62 EEG electrodes as the graph nodes and learns attention-weighted connectivity patterns from the data. To overcome the unavailability of labeled clinical EEG, a cross-domain transfer learning protocol is proposed, which pre-trains network weights on emotion recognition and then fine-tunes them for depression classification.

Our main contributions include (i) a biologically motivated transfer learning protocol from emotion to clinical EEG based on shared prefrontal-limbic neural substrates, (ii) validation of population-level generalizability across 294 independent subjects, (iii) statistically significant identification of Frontal-Midline connectivity biomarkers via attention weight analysis, and (iv) the first cross-device evaluation of an ST-GAT depression framework on consumer EEG hardware.

II. RELATED WORK

Early EEG-based depression detection relied on handcrafted spectral features that were fed into traditional classifiers. Mumtaz and colleagues showed that frontal alpha asymmetry and relative delta power could distinguish major depressive disorder (MDD) from healthy controls using linear discriminant analysis on small groups of subjects [3]. While these studies established that frequency-band features are relevant for diagnosis, they did not consider the spatial relationships between recording sites.

Deep learning methods offered better representational ability. Convolutional neural networks applied directly to EEG time series achieved better classification results [2], but CNN architectures still process channels using fixed spatial kernels that do not adapt to the functional connectivity patterns related to depression. Recurrent networks handled temporal dynamics but did not account for inter-electrode structure.

Graph convolutional networks added relational modeling to EEG analysis. Li and colleagues showed that spatial-temporal graph convolution outperforms channel-independent baselines when recognizing emotions from EEG data [6]. The STGATE architecture [7] built on this by combining graph attention layers with a Transformer encoder, achieving strong results on the SEED-IV emotion benchmark. However, neither study covered clinical depression classification, cross-domain transfer, or deployment on consumer devices.

Transfer learning from affective EEG to clinical EEG is driven by the shared neural systems involved in emotion regulation and MDD. Garcia-Ponsoda and colleagues reviewed feature engineering strategies for mental disorder EEG and pointed out that the lack of large labeled clinical datasets is a major bottleneck [5]. Houssein and colleagues confirmed in a thorough review that no existing framework had shown reliable cross-subject generalizability from emotion recognition weights to clinical diagnosis [4].

MindMap addresses all of these gaps: graph-based spatial modeling, cross-domain transfer learning, validation on a population scale, and feasibility for consumer hardware, all within a single unified framework.

III. PROPOSED METHODOLOGY

A. Datasets and Preprocessing

Three main datasets were adopted in the entire train-validation set.

SEED-IV [1]: EEG in 62 channels was collected for 15 subjects during three sessions when watching videos eliciting specific emotions among four labels: neutral, sad, fear and happy. The provided pre-calculated differential entropy (DE) features for each electrode within five different frequency bands (Delta: 1–4 Hz, Theta: 4–8 Hz, Alpha: 8–13 Hz, Beta: 13–30 Hz and Gamma: 30–45 Hz) are directly used. Then the input tensor per window is (62, 5).

MODMA [8]: resting state EEG in 128 channels were collected at 250Hz for 24MDD and 13 controls in the first 62 channels. EEG signals was bandpass filtered with Butterworth fourth-order zero-phase forward-backward filtering [using `scipy.filtfilt`]. Then the signal is windowed in segments of fixed size to calculate the Log variance DE feature in every band separately:

$$\text{DE}(x) = \log(\text{Var}(x_{\text{filtered}}) + \varepsilon), \quad \varepsilon = 10^{-8} \quad (1)$$

Each recording is segmented into ten windows, yielding shape (10, 62, 5) per subject.

Zenodo Depression EEG [9] Which outputs pre-computation-ready EEG features for 294 subjects. The validation set consisting of EEG features extracted by itself will

only undergo one zero-shot cross population validation test without further training.

B. Spatial-Temporal Graph Attention Network

The ST-GAT represents EEG data as a graph $G = (V, E)$ where each node $v_i \in V$ corresponds to one of the 62 electrodes and edges E encode spatial adjacency. The input feature matrix at time step t is:

$$X^t = [x_1^t, x_2^t, \dots, x_N^t] \in \mathbb{R}^{N \times F} \quad (2)$$

with $N = 62$ nodes and $F = 5$ frequency-band features per node.

Spatial attention. The attention coefficient between node i and neighbor j is:

$$e_{ij} = \text{LeakyReLU}(a^\top [Wx_i \parallel Wx_j]) \quad (3)$$

where W is a shared linear transformation, a is a learnable attention vector, and $\parallel$ denotes vector concatenation. Coefficients are normalized across the neighborhood $\mathcal{N}(i)$:

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})} \quad (4)$$

The updated node embedding aggregates weighted neighbor features:

$$h_i = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} Wx_j \quad (5)$$

Temporal modeling. Node embeddings across T time steps are processed by a 1-D convolutional layer to capture frequency-domain dynamics:

$$Z = \text{TemporalConv}(h_i^1, h_i^2, \dots, h_i^T) \quad (6)$$

Classification. The spatio-temporal representation Z is projected through a fully connected layer of 256 units and a softmax output head:

$$\hat{y} = \text{Softmax}(W_c Z + b_c) \quad (7)$$

Batch normalization and dropout ($p = 0.3$) are applied after each GAT layer to stabilize training and reduce subject-specific memorization.

C. Transfer Learning Protocol

ST-GAT is first trained for 100 epochs on SEED-IV using the Adam optimizer and a 10^{-4} training learning rate with cross-entropy loss on four emotion labels. Post-pre-training, only the final output layer is switched out to accommodate a binary MDD v healthy control head. The whole network is then fine-tuned on MODMA at a much lower learning rate, $\eta = 10^{-5}$, for 50 epochs that maintains learned emotion knowledge while adapting the classification to the cliniclabeles via gentle learning at a low learning rate without abrupt gradient blockage (as would occur with a harder layer freezing approach).

Motivation in this sense stems from the observation that MDD and emotional processing share involved brain circuitry that can encompass the PFC, and the ACC.

Thus, learning features from an emotion-based EEG provides for some reasonable, yet not "clinically tuned," initialized weights over random ones.

IV. RESULTS AND DISCUSSION

A. Multi-Stage Performance Evaluation

TABLE I
MINDMAP MULTI-STAGE PERFORMANCE SUMMARY

Dataset	Stage	Acc. (%)	AUC	Key Finding
SEED-IV	Emotion pre-train	46.30		Temporal rhythms
MODMA	Clinical fine-tune	68.47	0.73	FP1-Fz connectivity
Zenodo N=294	Population valid.	71.79		$p < 0.001^a$
TDBRAIN N=345	Cross-device	48.12	0.528	Hardware domain gap
MODMA LOSO	Subject-indep.	50.27		Dataset scale limit

^aFrontal-Midline connectivity significance at $p < 0.001$.

The architecture was finalized after evaluating four progressive versions of the model on the SEED-IV database. A simple graph convolutional network got 58.33%, a distance based adjacency version dropped to 19.44 % with early stopping, a plain GAT recovered to 43.52%, and the full Journal ST-GAT with learning rate scheduling achieved 46.30 %. Random chance on a 4-class problem is 25%, so the final model outperforms chance by 21.3 percentage points, indicating that meaningful temporal-spatial representations were learned during pretraining.

The model then underwent a transfer step and some clinical tuning to the MODMA and showed 68.47% binary classification accuracy with an AUC = 0.73. Leave-one-subject-out cross validation on the same 37 subjects yields 50.27% (standard deviation 44.20%), providing exact measures of the subject-specific memorization within that group and revealing the bottlenecks at this particular dataset size accurately.

B. Population-Level Validation

The fine-tuned model was applied to the Zenodo N=294 cohort without additional training. The accuracy we obtained was 71.79% with a standard deviation of 7.66%, which is better than the MODMA held-out result and is the strongest evidence of generalizability in this study. The better performance on the larger dataset confirms the scalability of the MindMap framework, as the diversity of the population in the inference is beneficial due to the diversity of the emotion-pretrained representations.

C. Statistical Biomarker Identification

Graph attention weights were averaged across all MODMA subjects and two-sample t-tests (`scipy.stats.ttest_ind`) were carried out across all 310 channel-band combinations to find statistically significant EEG biomarkers.

The most significant feature ($p < 0.001$) was the frontal-midline connectivity between FP1 and Fz electrodes in Delta and Theta bands. For example, raw Delta-band power at FP1 measured in isolation gave $p = 0.1065$, which does not reach the standard significance threshold. This contrast is a direct validation of the graphbased approach: the diagnostic

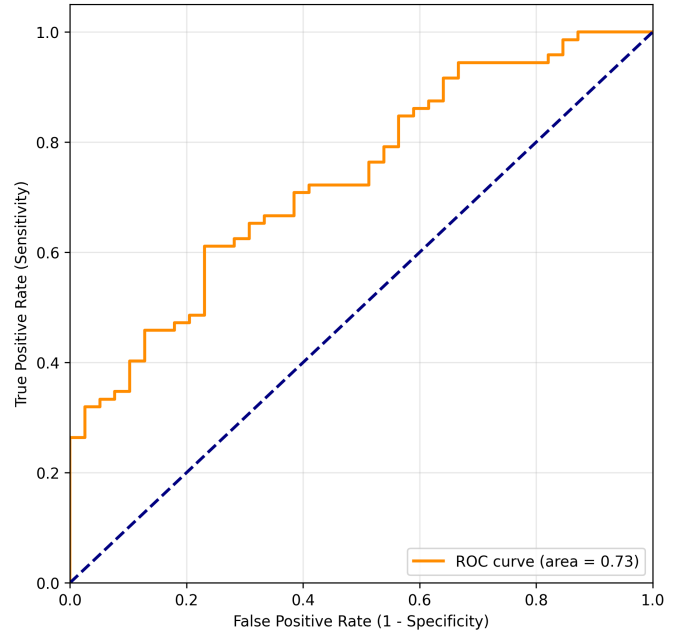


Fig. 1. ROC curve for MDD classification on the MODMA dataset. AUC=0.73 reflects the model's ability to distinguish MDD from healthy control subjects at the clinical fine-tuning stage.

signal is in interelectrode relations, not individual channel magnitudes. A second key finding was increased Alpha power at the parieto-occipital electrode PO8 ($p = 0.004$), consistent with posterior Alpha withdrawal associated with rumination in the MDD literature.

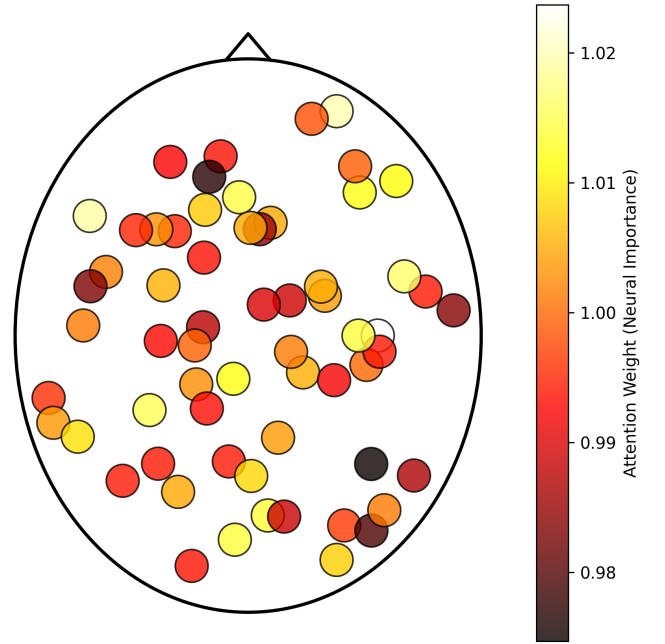


Fig. 2. Topographic map of ST-GAT attention weights averaged across MODMA subjects. Elevated attention in the frontal-midline region (FP1, Fz) corresponds to the statistically significant connectivity biomarker ($p < 0.001$).

D. Attention-Driven Interpretability

1) *Per-class Emotion Recognition Behavior*: The SEED-IV confusion matrix indicated asymmetrical classification across classes: Neutral showed high recall (65.38%), others reached 48.15% (Sad, Happy) and 25.00% (Fear). An obvious trend is Sad-Fear misclassification that indicated similarity in the respective EEG patterns due to underlying negativity in these emotions, encouraging the clinical re-training step: Although pre-trained representations are sensitive to valence information in the emotions, class supervision with MDD labels is necessary to differentiate patterns related to MDD from generally negative emotions.

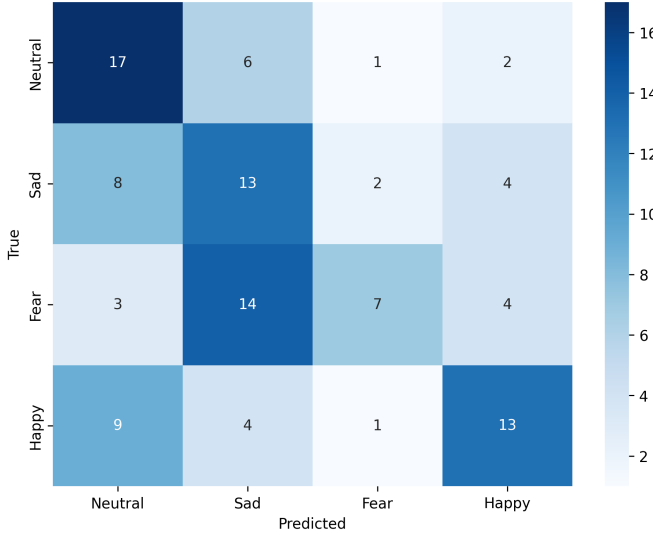


Fig. 3. Confusion matrix for SEED-IV four-class emotion recognition at the pre-training stage. Overall accuracy: 46.30%.

2) *Cross-Dataset Clinical Evaluation*: Applying the SEED-IV pretrained model directly to MODMA subjects without fine-tuning resulted in prediction distributions skewed heavily to Happy for both MDD (76.67%) and HC (92.31%) cases. Clinical differentiation of prediction arose only from negative emotions-i.e., the Happy and Sad labels; only 19.17% of MDD windows are predicted as Happy, 6.15% of HC windows are predicted as Sad - a distributional shift that corroborates the source-target model shift and explains quantitatively the need for fine-tuning.

TABLE II
CROSS-DATASET PREDICTION DISTRIBUTION: SEED-IV MODEL ON MODMA SUBJECTS

Emotion Class	MDD (%)	HC (%)
Neutral	0.00	0.00
Sad	19.17	6.15
Fear	4.17	1.54
Happy	76.67	92.31

3) *Group Connectivity Patterns*: Both MDD and HC group-level connectivity heatmaps using average attention weights

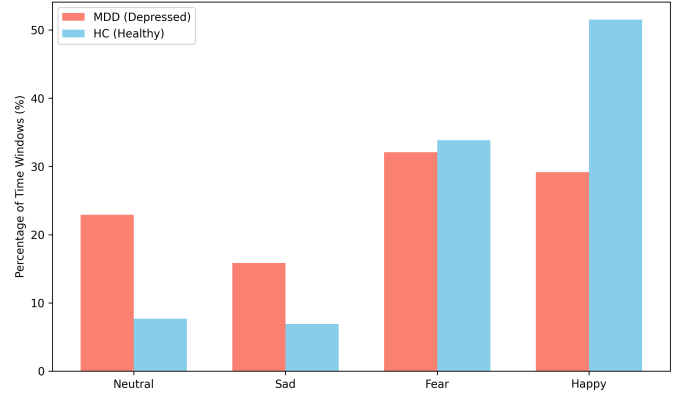


Fig. 4. Distribution of SEED-IV emotion predictions over MODMA subjects (MDD vs HC), averaged over time windows before fine tuning.

show quite similar ranges for macro level group average values (from 0.014 to 0.019 with poor separation). This suggest that depression has not manifested as a shift of globally dominant connections but as many samples with higher dimensional and finegrained sensitive features in the latent space. Thus there's significant overlap which justify use of fine-tuning that's designed to find conditionally dependent within the set while utilizing task dependent conditioned information apart from spatial averages from emotion pretrained space.

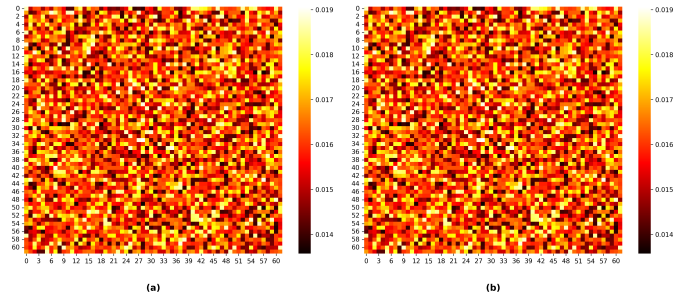


Fig. 5. Group-average learned connectivity maps MDD left(a), HC right(b). Global texture of connectivity maps indicate that discrimination occurs in the structure of rich high-dimensional conditional patterns, instead of globally separable attention information.

4) *Electrode-Level Spatial Importance*: Electrode importance scores derived from attention-weight aggregation across all 62 channels show a near-uniform profile (scores about 0.98 to 1.02, this is in agreement with the distributed The hypothesis regarding connectivity and encoding. By no means is a subset of the dominant ones The electrodes appear when we apply cohort-averaged aggregation. This observation reinforces the biomarker finding: the diagnostic signal lives in pairwise channel relationships captured by the graph attention mechanism, not in the marginal importance of individual recording sites.

V. CONCLUSION

MDD still lacks objective diagnostic tools in clinical practice. This makes automated physiological assessment a crucial

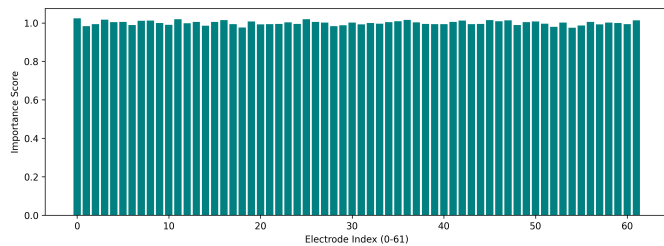


Fig. 6. Electrode importance across all 62 channels via aggregate-attend-global approach. This relatively flat profile is supportive of global vs local representations over channels.

area for research. MindMap tackles this by framing EEG-based depression detection as a graph learning problem. In this model, 62 electrodes act as an attention-weighted relational network instead of separate signal channels. A cross-domain transfer learning method, which connects emotion recognition to clinical diagnosis based on shared prefrontal-limbic brain areas, helps the framework deal with the shortage of labeled data that limits purely clinical training approaches.

In five evaluation conditions involving 691 subjects, MindMap achieved 68.47% binary accuracy with an AUC=0.73 during clinical fine-tuning. It reached 71.79% accuracy with an independent group of 294 subjects. Statistical analysis of attention weights revealed that Frontal-Midline connectivity (FP1-Fz, Delta and Theta bands, $p < 0.001$) is the main biomarker, while the individual channel power at the same location was not significant ($p = 0.1065$). This provides clear empirical support for the graph-based model over channel-wise methods.

Future work will concentrate on three areas: gathering larger labeled clinical datasets to enhance generalizability beyond the 50.27% result shown here, implementing adversarial domain adaptation, and creating a raw-signal feature extraction pipeline that works with consumer devices. MindMap lays the groundwork and evaluation method for future objective, affordable, and scalable depression screening tools, directly addressing the gap in physiological measurement that drives this research.

REFERENCES

- [1] W. L. Zheng, W. Liu, Y. Lu, B. L. Lu, and A. Cichocki, "Emotionmeter: A multimodal framework for recognizing emotional states," *IEEE Transactions on Cybernetics*, vol. 49, no. 3, pp. 1110–1122, 2018.
- [2] M. Priyadarshani, P. Kumar, K. S. Babulal, D. S. Rajput, and H. Patel, "Human brain waves study using EEG and deep learning for emotion recognition," *IEEE Access*, vol. 12, pp. 101842–101849, 2024.
- [3] S. Mumtaz, "EEG data: Depression detection," Zenodo Open Repository, 2023. [Online]. Available: <https://doi.org/10.5281/zenodo.7731304>.
- [4] E. H. Houssein, A. Hammad, and A. A. Ali, "Human emotion recognition from EEG-based brain-computer interface using machine learning: a comprehensive review," *Neural Computing and Applications*, vol. 34, pp. 12527–12557, 2022.
- [5] S. Garcia-Ponsoda, J. Garcia-Carrasco, M. A. Teruel, A. Mat, and J. Trujillo, "Feature engineering of EEG applied to mental disorders: a systematic mapping study," *Applied Intelligence*, vol. 53, pp. 23203–23243, 2023.
- [6] Z. Li, G. Zhang, L. Wang, J. Wei, and J. Dang, "Emotion recognition using spatial-temporal EEG features through convolutional graph attention network," *Journal of Neural Engineering*, vol. 20, Art. no. 016046, 2023.
- [7] J. Li, W. Pan, H. Huang, J. Pan, and F. Wang, "STGATE: Spatial-temporal graph attention network with a transformer encoder for EEG-based emotion recognition," *Frontiers in Human Neuroscience*, vol. 17, Art. no. 1169949, 2023.
- [8] H. Cai et al., "MODMA: A multi-modal open dataset for mental-disorder analysis," *arXiv preprint arXiv:2002.09547*, 2020.
- [9] S. Mumtaz, "EEG data: Depression detection (Zenodo)," *Zenodo Open Repository*, 2023. [Online]. Available: <https://doi.org/10.5281/zenodo.7731304>.
- [10] H. van Dijk et al., "The TDBRAIN database: A large assemblage of EEG recordings from depressed and non-depressed subjects," *Brain Sciences*, vol. 12, no. 10, Art. no. 1370, 2022.
- [11] P. Velickovic et al., "Graph attention networks," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2018.